

Zihan Fang

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Research Focus

My research develops human-centered AI to lower barriers to acquiring expertise and contributing effectively to software development. I combine large-scale repository mining, cognitive and behavioral modeling, and reinforcement learning to identify sources of difficulty, characterize how developers acquire expertise, and translate these insights into AI-supported programming systems and immersive learning environments that improve learning efficiency, programming performance, and long-term skill development.

Education

Vanderbilt University, Ph.D. in Computer Science	Aug 2023 – Present
Vanderbilt University, M.S. in Data Science; GPA: 3.9/4.0	Aug 2021 – May 2023
Soochow University, B.S. in Information Management; GPA: 3.8/4.0	Sep 2016 – Jun 2020

Publications

- [1] **Zihan Fang**, Janice Chung, Ruijia Chen, Marcia Barnes, Yuhang Zhao, Yu Huang: *Toward Inclusive Programming Support for Novice Programmers with Dyslexia: Insights from a Comparative Study*. In the *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*.
- [2] Ningzhi Tang, Chaoran Chen, **Zihan Fang**, Gelei Xu, Maria Dhakal, Yiyu Shi, Collin McMillan, Yu Huang, Toby Jia-Jun Li: *Programming by Chat: A Large-Scale Behavioral Analysis of 11,579 Real-World AI-Assisted IDE Sessions*. In the *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*.
- [3] **Zihan Fang**, Weizhe Jiao, Robert Tairas, Yu Huang: *Novices Comprehending Functional vs. Imperative Programs: An Eye-Tracking Study*. *ACM Transactions on Software Engineering and Methodology (TOSEM)*.
- [4] Yueke Zhang, **Zihan Fang**, Kevin Leach: *Stop the Retrieval Thrash: Brain-Guided Episodic Memory for Repository-Scale Coding Agents*. In *Companion Proceedings of the ACM on Software Engineering*, Vol. 3, *FSE (FSE 2026, HumanAISE)*.
- [5] Yueke Zhang, Yifan Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs for Code*. In *Proceedings of the ACM on Software Engineering*, Vol. 3, *FSE (FSE 2026)*.
- [6] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *Locating Software Vulnerabilities With Static Analyzers: How Far Are We?* In *Proceedings of the 30th International Conference on Evaluation and Assessment in Software Engineering (EASE 2026)*.
- [7] **Zihan Fang**, Robert Wallace, Zachary Karas, Toby Jia-Jun Li, Collin McMillan, Yu Huang: *Investigating the Feasibility of Conducting Webcam-Based Eye-Tracking Studies in Code Comprehension*. *IEEE Transactions on Software Engineering (TSE)*.
- [8] **Zihan Fang**, Yueke Zhang, Thomas Zimmermann, Denae Ford, Yu Huang: *Contribution Patterns in Open Source Software for Social Good: Dynamics, Individuals, and Impact*. In *Proceedings of the ACM on Human-Computer Interaction (CSCW 2026)*.
- [9] Yueke Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *CodeACT-R: A Cognitive Simulation Framework for Human Attention in Code Reading*. In *Companion Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025, NIER Track)*.
- [10] **Zihan Fang**, Jiliang Li, Anda Liang, Gina R. Bai, Yu Huang: *A Comparative Study on ChatGPT and Checklist as Support Tools for Unit Testing Education*. In *Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (FSE 2025, SEET Track)*.
- [11] **Zihan Fang**, Yu Huang: *“Math Is a Pain!”: Understanding Challenges and Needs of the Machine Learning Community on Stack Overflow*. In *Proceedings of the ACM on Human-Computer Interaction*. (*CSCW 2024*)
- [12] **Zihan Fang**, Madeline Endres, Thomas Zimmermann, Denae Ford, Westley Weimer, Kevin Leach, Yu Huang: *A Four-Year Study of Student Contributions to OSS vs. OSS4SG with a Lightweight Intervention*. In *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023, ACM SIGSOFT Distinguished Paper Award.)*

Manuscripts Under Review

- [1] **Zihan Fang**, Yifan Zhang, Yueke Zhang, Kevin Leach, Yu Huang: *DPO-F+: Aligning Code Repair Feedback with Developers’ Preferences*. [arXiv]
- [2] **Zihan Fang**, Yueke Zhang, Yu Huang: *AgentForge: An Immersive Role-Playing Platform for Learning Agentic Software Engineering*. [arXiv]

[3] **Zihan Fang**, Yueke Zhang, Ningzhi Tang, Collin McMillan, Toby Jia-Jun Li, Yu Huang: *From Conversation to Contribution: Characterizing Coding Agent in Open-Source Software*. [arXiv]

[4] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *GraphAlignCoder: Aligning Program and Proof Graphs for Code Generation*. [arXiv]

Research and Industry Experience

Research Assistant , Vanderbilt University	Aug 2023 – Present
• Conduct research on human-centered AI for software engineering, spanning developer-feedback alignment, AI-assisted programming, and multi-agent learning environments.	
Data Scientist , Tencent Inc.	Aug 2021 – Dec 2021
• Built scalable ML and data systems to support real-time identity verification and model development.	
Data Analyst Intern , DiDi Inc.	Jan 2021 – Jun 2021
• Applied data analytics and machine learning to segment drivers, evaluate operational strategies, and optimize recruiting, onboarding, and business development efforts.	

Teaching Experience

Teaching Assistant , Vanderbilt University	Aug 2022 – Present
• CS3892/5892 Designing Computing for Learning – Fall 2026 • DS3100 Fundamentals of Data Science – Fall 2022, Spring 2023 • CS3265 Database Management Systems – Fall 2023 • CS4278/5278 Principles of Software Engineering – Spring 2024	

Awards

• SIGSOFT CAPS ASE Travel Award	2026
• Ignite Grant at Vanderbilt University	2025
• LIVE Horizon Grant at Vanderbilt University	2025
• C. F. Chen Best Paper Award in the Department of Computer Science at Vanderbilt University	2024
• Richard Bennett/Dorothy Danforth Compton Prize Scholarship at Vanderbilt University	2024
• SIGSOFT Distinguished Paper Award at FSE	2023
• NSF Student Travel Award	2023

Service

AAAI 2026: PC Member

Empirical Software Engineering (EMSE): Reviewer

SIGCSE TS 2025: PC Member (Posters; Computing Education Research)